S6 Internal Changelog

Internal analysis provenance and manuscript revision history

Article type: Supplement

Short title: S6 Changelog - NDT v2

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Scope and intent

Internal working document for the NDT manuscript rebuild. Public claim discipline lives in the manuscript; this file records drafting provenance.

Provenance

Sciencelead freeze: `sciencelead/sl_1.md`, `sciencelead/sl_001.md`. Source intake: `summaries/00_synthesis.md` and `summaries/01–06`. Identity donor: `articles/NDT/NoeticDiffusionTheory.typ`. Epistemic donor: `articles/NDT_2/ndt_2.typ`.

Changelog

2026-08-28 — first prose pass (manuscript §§1–8)

Replaced the generic empirical template with theoretical-framework architecture. Wrote scope, introduction (state → accessibility), recoverability ladder, canonical SDE with non-conservative split, MNPS/NMD boundary, local Jacobian summaries, diffusion tensor, and possible-futures objects (reachability, committor, FAR). Control, empirical anchors, and neighbouring-theory comparison are outlined but not yet drafted. Companion experiment IDs are kept out of the public text; DANDI 000458 appears only as the Level-1 / Level-4 illustration already used in ndt_2.

2026-08-28 — control layer (manuscript § Biological control)

Drafted RVC, thalamic access, HELS vs HELS-A vs HELS-A1, embodiment/metabolism, and dendritic M_b. HELS-A intake from `ndt-subsystems/project/HELS-A` (H0/H0B eligibility; M0/M0R fabricated-estimator FAIL; A1 and spikes closed) plus simulator A0/A0B operator existence on frozen Φ. Empirical anchors and neighbouring theories remain stubs.

2026-08-28 — empirical anchors (manuscript § Empirical constraints)

Replaced the empirical stub with three main-text anchors (spindle-locked d_n timing; I-CARE recovery-linked reachability/Jacobian; DANDI 000458 Level-1 / Level-4 split) plus a compact dissociation matrix. Neighbouring theories remain a stub. Dataset records added to `references.bib`.

2026-08-28 — neighbouring frameworks and predictions

Replaced the remaining stub with neighbouring frameworks (positioning, dynamical family, bridge table, Cogitate / state-vs-experience discipline) and predictions (P1–P7 plus hierarchical failure). Source harvest: `J:/repos/NoeticDiffusion/articles/ndt_compared_to_other_theories`, rewritten for NDT 1.0 (accessibility, not denoising identity).

2026-08-28 — discussion, limitations, conclusion

Closed the main text: discussion (denoising → accessibility; retain / promote / separate; consciousness-research implication), limitations, and conclusion thesis from `sciencelead/sl_001.md` §§18–20. Public supplements remain template stubs.

2026-08-28 — conceptual figures (script + five sciencelead panels)

Added reusable generator `src/make_conceptual_figures.py` (PNG+SVG, CLI `--only` / `--formats` / `--list`). Emitted fig1–fig5 into `figures/conceptual/` and wired them into the manuscript. Figure 5 cell codes follow summaries `01`/`03`/`04`; anesthesia and epilepsy rows are unpaid.

2026-08-28 — canonical coma freeze, S1–S5 populated

Audited pr_1 comment 7 against `articles/CLINICAL/comatose_ndt/Recovery-Linked Neural Geometry After Cardiac-Arrest Coma.typ`. The 298-patient ComBat merge is an intermediate; the canonical article

is `20260606_combined_invalidityfix` (301/281, AUROC $0.599 \rightarrow 0.662$). The article now binds that source (S2) and says patients, not recordings. Claim-ledger HELS/seed language moved toward S3. S1–S5 filled (math/kernel, source freeze, negatives, clinical translation, glossary).

2026-08-29 — pr_2 core: kernel in main text

Moved $\mathcal{L}$ and protocol-conditional $K^{\gamma}_{t:t+h}$ plus revised C1–C3 into the main paper as a transition-law section after the canonical SDE. Drift split now uses mobility M and potential V (IIT Φ untouched); main text no longer claims that pure potential descent cannot represent reactivity. Coordinate-dependence warning, predictive-spread ellipsoid terminology, and renamed non-equivalent questions are in the main text. Biological-control section cut to three classes. Companion analyses cited alongside datasets. P8 within-state experience test added. S3 dropped the closure-signature negative. S6 remains internal and is not a journal supplement.

2026-08-30 — S4 rebuilt as non-technical explanation

Replaced the clinician-facing S4 with a plain-language companion (`S4_Non_technical_explaination.typ`). Tone and metaphors are borrowed from the **Reality from Noise** essays (waking disorientation, map versus land, texture versus elevation, ridge in fog). Claims stay on NDT 1.0: accessibility as a future law, MNPS as a chart, RVC as a special case, three anchors as constraints. The older denoising identity is explicitly not restored. Main text points to S4 after the noetic-accessibility paragraph.

2026-08-30 — public tone: flagship / programme

Replaced internal labels (**flagship**, self-referential **programme**) in the main text and S1–S5 with **theory**, **article**, or **present theory**. Neighbouring empirical programmes (HOT/AST) left as ordinary academic usage. S6 remains internal.

2026-08-30 — pr_3: m3-bridge targeted import, then freeze

External Science Lead review (`peer_review/pr_3.md`) plus summaries `07`–`10`. Empirical intake for this draft is now frozen. Three anchors unchanged. Kernel $\mathcal{L}$ / K^h, mobility M, potential V, weaker drift-split wording, and Jacobian coordinate dependence were already in the main text from pr_2; they were not re-litigated.

- Anchor 3 cites the m3-bridge 000458 article of record: $r = 0.406$ / $r = 0.395$, residual 0.856, alignment incomplete.
- Recoverability ladder elevated as a methodological contribution, with IBL decoder PASS / discrete-flow STOP as a second worked boundary (not Anchor 4).
- TRN section: joint TRN $\rightarrow$ meso $\rightarrow$ surface-macro chain not estimable from audited archives.
- Figure 5 / dissociation table: AVB state geometry, IBL readout STOP, DANDI 000574 human MTL–scalp null. ARR, AGR, FB3, 24h, and Cecchetto stay out of the main matrix.
- S1/S3: ARR condition-mixing inflation of $\mathrm{Tr}(Q)$ in 126/126 fits; 000458 three-stage chain blocked. S5: chart-speed coupling $\neq$ stochastic reachability. S2 source freeze updated. S6 remains internal.

2026-08-30 — Zenodo DOI refresh

Public records for ORCID `0009-0006-6927-7456` (59 hits). Companion DOIs updated where a public version exists:

- NDT 0.8 identity paper remains `10.5281/zenodo.20112950` (latest of concept `17367307`).
- Coma companion: `10.5281/zenodo.21263084` (v.0.3), not v.0.1.1 `20410486`. Analysis freeze unchanged.
- Anchor 3: `10.5281/zenodo.22041386` (v.0.1).
- Neighbourhood essay: `10.5281/zenodo.21226430`.
- Still unpublished: RVC three-dataset spindle companion, IBL2 discrete-flow STOP, AVB omission, DANDI 000574.

2026-08-31 — pr_4 literature neighbourhood

Abstract already rewritten by the author; not retouched. Author Summary kept as the measurement-facing companion.

Introduction now opens from population-dynamics literature (Breakspear, Cunningham–Yu, Vyas, Duncker, Hancock, Hudson, Barttfeld) before NDT history. Added 17 neighbourhood citations: non-normal transients (Hennequin, Christodoulou), TPT (E–Vanden-Eijnden), thalamic access (Halassa, Schmitt), cortical gain (Ferguson), context-dependent dynamics (Mante), variability quenching (Churchland), recoverability (Nejatbakhsh), and consciousness-dynamics repertoire (Demertzi). No new empirical anchors.

2026-08-31 — pr_5: present-tense theory, not a version diff

Removed v0.8→v1.0 narration from the public article. Author Summary no longer says **rebuilds**. Introduction presents RVC/MNDM/MNPS as a restricted realization of accessibility control. RVC is derived as the isotropic variance-control limit. Thalamic prose drops **obsolete slogan**. Discussion 13.1/13.2 collapsed into **Dynamical accessibility as the organizing principle**, with one continuity citation to the earlier formulation. Release-note detail stays in this internal S6 / Zenodo version notes.

2026-08-31 — pr_6: journal-surface cleanup

Architecture frozen. Main-text cleanup only: repository/governance phrasing translated (no **fail closed**, **product-level export**, **source freeze**, **unpaid**, PASS/STOP in captions). NMD section cut to the measurement-contract distinction. Diffusion SDE given an explicit non-Brownian / non-Markov scope sentence. C2 renamed **non-redundant accessibility**. Empirical coma freeze hash moved to S2. Defensive reviewer sentence in neighbours replaced. Conclusion left intact.

2026-08-31 — pr_7: kernel split and compression

Architecture remains frozen. Formal fix: latent kernel K^X versus chart kernel K^x, with dummy ξ in the generator; unadorned K^h means K^x. C1/C3 and P8 follow the chart-conditional object. DANDI 000458 / IBL numbers live only in Anchor 3; §3 is a compact claim-level example. Coma ceiling says study-specific chart-level estimators. NMD wording is specifies / operationalizes. Generator uses $\frac{1}{2}$. Light §9 citations added for truth-known, TRN geometry, and Slow Branch where public DOIs exist; HELS-A and the two-operator simulator still point to S3. No fourth anchor. S6 remains internal.

2026-08-31 — pr_8: S4 conceptual decoder and figures

S4 reorganized as a pedagogical derivation (same place / several questions / map vs land / matched-state control / structure vs access). Softened the older-essay sentence and the closing line so they do not read as a version diff. Five sparse S4 schematics added to `src/make_conceptual_figures.py` (`s4a`–`s4e`); not a single packed infographic. Main text gains a hiker primer box after Figure 1. S6 remains internal.

2026-08-31 — pr_9: C1/C3 non-triviality and package consistency

C1 alone is not distinctive; empirical content is C2/C3. C3 is not generic covariate conditioning. Latent ξ restored in a and the drift split; chart estimates marked $\hat{b}_x$ etc. Named summaries are non-equivalent functionals, not projections of one kernel. Evidence class and inferential claim class declared orthogonal; “ladder”/“climb” language removed. S4 hide-cites Langell 2026 keys in main-article order so APA letters match (then a–i; now a–k after the 2026-09-01 DOI refresh). S4 retitled as conceptual guide; Jacobian, hippocampal, autonomic, and “two minds” wording tightened. No new figure, dataset, or mechanism.

2026-09-01 — Zenodo DOI refresh (spindles, TA simulator, Rolón)

Public records for ORCID `0009-0006-6927-7456` (63 hits). Companion DOIs updated where a public version exists:

- Spindle companion: `10.5281/zenodo.22206108` (v.0.1), two-dataset title. Not the earlier window paper `20038829`.
- Two-operator simulator: `10.5281/zenodo.22204100` (v.0.1), now cited in §9 / S3 instead of “remain in S3”.
- Rolón-Martínez covariance-shape: `10.5281/zenodo.22229790` (v.0.1). This is the new empirical TA deposit, not DANDI 000363.
- Still unpublished: IBL2 discrete-flow STOP, AVB omission, DANDI 000574. DANDI 000363 has no manuscript and no public DOI.

Main-article Langell 2026 first-cite order is now a–k (simulator e, TRN-VG f, Rolón g, Slow Branch h, spindles i, coma j, neighbourhood k). No fourth empirical anchor. S6 remains internal.

2026-09-02 — IBL2, AVB, DANDI 000574 DOIs

`python -m zenodo catalog` wrote `zenodo/langell_articles.json` (67 public records). Companion DOIs:

- IBL2: `10.5281/zenodo.22231817` (v1.0).
- AVB omission: `10.5281/zenodo.22232650` (v1.0).
- DANDI 000574: `10.5281/zenodo.22243202` (v1.0).
- Empirical TA companion remains Rolón-Martínez `10.5281/zenodo.22229790`, not DANDI 000363.

First-cite order is now a–m (AVB k, 000574 l, neighbourhood m). No fourth empirical anchor. S6 remains internal.

2026-09-02 — pr_10 visual / notation freeze pass

Architecture remains frozen. Notation: empirical Jacobian $\hat{J}_x$; P6 uses measured x_t. Accessibility kinds $\mathcal{L}$ / K^h / q / FAR distinguished. Dendritic M_b shortened in §9; detail in S3. Coma prose uses predictive-spread summaries. Novelty: systematically and state-dependently modulated; synthesis not new stochastic mathematics. Fig 2 enlarged; Fig 5 header separated from title; Fig 4 given a weak page break. DANDI 000458 dataset citation completed. Claim-class wording replaces leftover Level 1/4 in public text. Funding / competing-interest placeholders resolved. No new dataset, mechanism, or theoretical layer.

2026-09-02 — HELS-A simulator DOI

HELS-A companion: `10.5281/zenodo.22253543` (v.0.1), **Context-Conditioned Access Operators Reshape Route Accessibility on Fixed Dynamical Landscapes**. Cited in §9 and S3. First-cite order is now a–n (HELS-A h, Slow Branch i, spindles j, coma k, AVB l, 000574 m, neighbourhood n). Not a fourth empirical anchor. S6 remains internal.